

Integrating Multi-Attribute Utility Theory and Sales-Response Optimisation to Develop a Data-Driven Doctor Targeting Model for Pharmaceutical Launch Planning

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Abstract

Introduction: Pharmaceutical product launches in regional markets are constrained by a structural mismatch between the size of the prescriber universe and the field resources available to engage it. Conventional planning distributes medical representative effort and promotional budget uniformly across all doctors, an approach that produces contact frequencies and per-doctor investment levels too low to influence prescribing behaviour at any point in the distribution.

Aim and Objectives: This study aimed to develop and validate a quantitative decision model for prescriber segmentation and resource allocation under simultaneous constraints of field capacity, promotional budget and launch time horizon, using an integrated approach combining multi-attribute utility theory, sales-response optimisation, scheduling heuristics and concentration analysis.

Methods: A composite Doctor Value Score (DVS) was formulated as a weighted additive value function over five pillars—prescription potential (0.30), adoption propensity (0.25), influence and network power (0.20), accessibility (0.15) and competitive vulnerability (0.10)—each scored on anchored 0–10 rubrics. Validity was established under the mutual preferential independence condition of Keeney and Raiffa. Visit frequencies were derived from the concavity of detailing response by Lagrangian optimisation under a binding capacity constraint. Territory allocation was formulated as makespan minimisation on identical parallel machines and solved using the Longest Processing Time heuristic. Value concentration was quantified using Lorenz curves and the Gini coefficient. All model outputs were verified against an independent computational audit.

Results: The model was applied to a representative universe of 600 prescribers served by six medical representatives with a quarterly promotional budget of ₹30 lakhs. Composite scoring stratified the universe into four tiers (Platinum ≥ 8.0 , Gold 6.0–7.9, Silver 4.0–5.9, Bronze < 4.0), identifying approximately 200 prescribers (33%) projected to contribute the majority of prescription volume. Derived call frequencies of 3.5, 2.0, 1.0 and 0.33 visits per prescriber per month generated a demand of 750 calls monthly against a capacity of 1,188, corresponding to 63% field utilisation and confirming operational feasibility; the plan remained deliverable at 76% utilisation under a pessimistic capacity assumption. Value-proportional budgeting achieved a 4.15-fold concentration of per-prescriber investment in the highest tier relative to uniform allocation, and territory balancing produced a workload coefficient of variation of 1.8%, within the 10% tolerance accepted in practice. Independent recomputation of 123 numerical assertions verified 85 exactly and identified six material errors, including a cumulative breakeven point overstated by five to six months and a return-on-investment figure misreported as a revenue-to-cost ratio.

Conclusion: The findings demonstrate that prescriber targeting and promotional resource allocation can be formulated as a tractable constrained optimisation problem rather than a matter of managerial judgement, yielding an auditable and operationally feasible allocation plan. The model is implemented as a functional decision-support application and provides a reproducible basis for empirical recalibration once observed prescribing data become available.

Keywords: Multi-attribute utility theory, doctor value score, pharmaceutical sales planning, resource allocation, sales-response optimisation, network segmentation